

Module-4 Individual Task-4

Artificial Neural Network (ANN)

Introduction to ANN

An Artificial Neural Network (ANN) is a machine learning model inspired by the human brain. It consists of interconnected nodes called neurons.

An ANN has three main layers:

- Input Layer
- Hidden Layer
- Output Layer

In a feedforward neural network, data moves only in one direction: from input to output.

Structure of Neural Network

The neural network contains:

- **Input Layer (X_1, X_2, X_3)** → takes input data
- **Hidden Layer (H_1, H_2, H_3)** → processes data
- **Output Layer (Y)** → gives final result

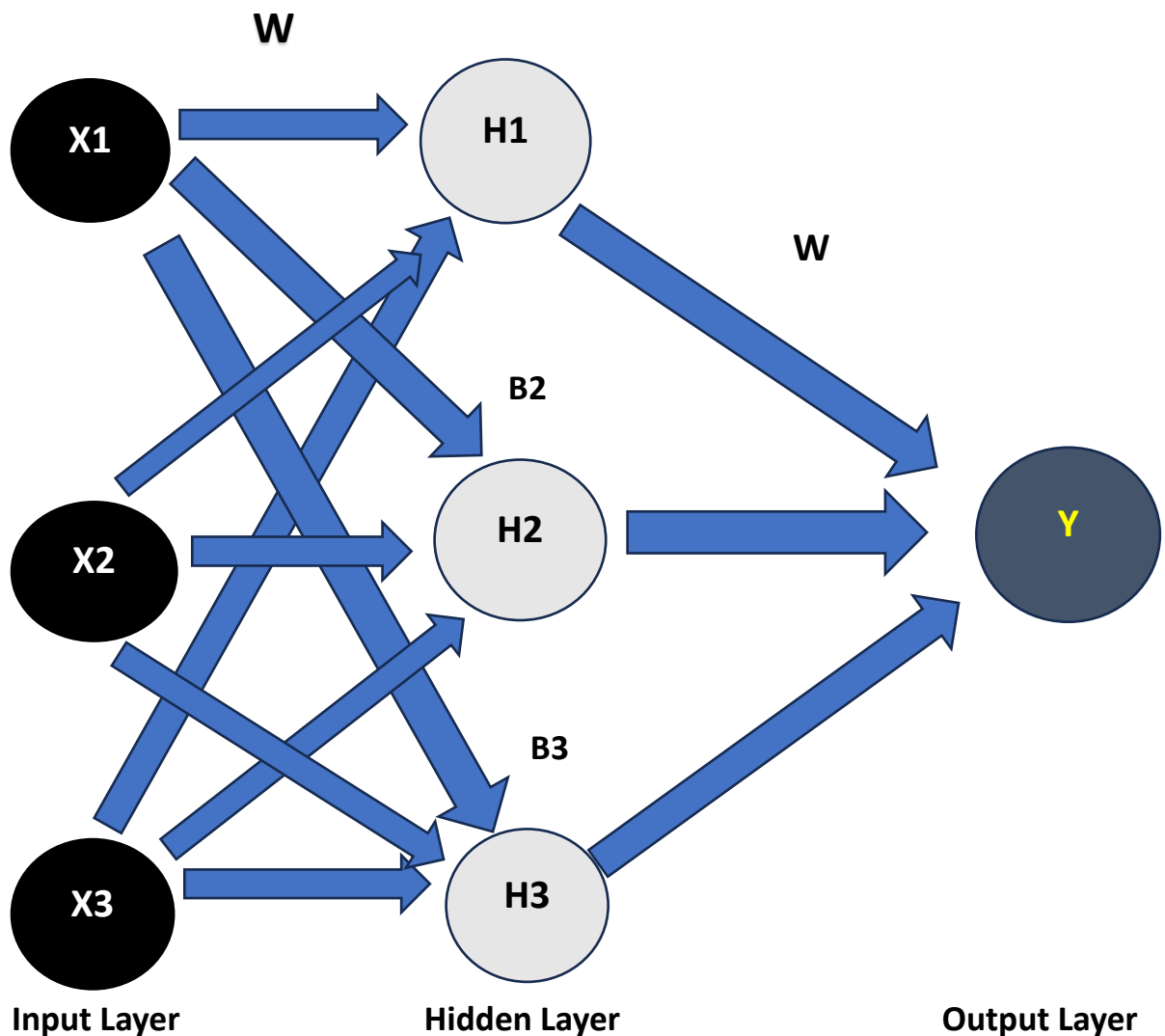
Each connection has:

- **Weights (W)** → importance of input
- **Bias (b)** → extra value added to neuron
- **Activation Function** → decides output

Structure(Diagram)

W=Weight

B1



$$Z=(W1*X1)+(W2*X2)+(W3*X3)+B$$

Activation Functions

The result Z is passed through an activation function to decide the neuron's output. This allows the network to learn complex, non-linear patterns

Key Formulas:

- ReLU: $f(z) = \max(0, z)$
- Sigmoid: $f(z) = 1 / (1 + e^{-z})$

Training and Error Calculation

Making a Prediction

In a feedforward neural network, data moves from the input layer through the hidden layers to the output layer. This process is called Forward Propagation. The network generates a prediction based on the current weights.

The Loss Function

To improve, the network must know how wrong its prediction was. We use a Loss Function to calculate the difference between the Actual Result and the Predicted Result.

The Error Formula

$\text{Loss} = (\text{Actual} - \text{Predicted})^2$

Gradient Descent

Gradient Descent is an optimization algorithm used to minimize the Loss. It calculates the "slope" of the error to determine which direction to adjust the weights to make the model more accurate.

Backpropagation and Learning

Backpropagation

This is the most important part of the learning process. The network moves backward from the output layer to the input layer to see which neurons caused the error. It uses the **Chain Rule** from calculus to find the gradient for every weight.

Updating the Weights

The network updates each weight by subtracting a small portion of the gradient. This is controlled by the **Learning Rate**.

The Update Formula

New Weight = Old Weight - (Learning Rate * Gradient)

Training Epochs

The network repeats these steps (Forward Pass -> Loss -> Backpropagation) thousands of times. Each full pass through the dataset is called an **Epoch**. Eventually, the weights become optimal, and the network can predict results accurately.

Conclusion

Artificial Neural Networks use these mathematical cycles to simulate human learning. By constantly adjusting weights based on error, they can solve complex problems like image recognition and data classification.